

Logix 1400 has various digital input and digital output pins operated by +24V DC. An analogue input module is added separately. Another high-current 1000A power supply device is connected via RS232 protocol, where only three lines, namely,

TX, RX and GND, are used for communication via a PLC serial port. Finally, AB PLC is connected to the PC via an Ethernet port.

RS232. RS232 is a serial protocol that converts parallel data to serial bits (pulses) and sends these across three wires, that is, TX, RX and GND. In most cases, the number of wires required is just three, but in special circumstances it may go up to nine wires, say, an RS232 modem.

Allen Bradley 1400 series has two serial ports. The first one is through a PPI cable with a 9-pin D connector at the other end. This needs a NULL modem connector to connect to any PC having a serial port or USB serial adaptor. NULL modem connections are shown in Fig. 4.

Interfacing the PLC to a computer. MAX1400 has three ports. First is COM1 (round connector); second is COM2 (9-pin D connector), which is for RS485/

RS232 communications; and the third is Ethernet for Ethernet/IP driver based communications. Therefore it is pretty easy for anyone to configure Ethernet as download port and COM1 as serial port for RS232 communications.

Allen Bradley PLC is connected to the computer via the Ethernet. We need to interface the programmable controller in order to configure and program (Fig. 6).

Adding drivers. Adding a driver is required for RSLOGIX classic software. To do so, you need to click on the middle icon and add RS232 DF1 driver to it.

Software for interfacing. RSLinx Classic is a software tool from Rockwell Automation Networks and Devices. It is a comprehensive factory communication solution, providing Allen Bradley PLC access to a wide variety of Rockwell Software and Allen Bradley applications, ranging from device programming and configuration applications such as RSLogix.

BOOTP server is a suitable server to establish communication between PC and Rockwell PLC.

Communicating to hyper terminal. How does one know that the PLC is transmitting characters out of its serial port? You can view the pulses or protocol produced using a hyper terminal window.

For that, you need to set the hyper terminal settings to 19200, N, 8, 1, which is same as the settings of PLC serial DF1 driver.

Double-click MAC and enter the IP (Fig. 7).

Ladder programming. Once the AB PLC is successfully interfaced to the PC, we need to properly program it using RSLOGIX provided by Rockwell Automation. For programming the PLC, we use Ladder programming. Typical PLC programs use blocks like timers on and off, counters, digital I/Os, RS232 communication blocks and others.

Ladder program is similar to the wiring diagram of the circuit. We redraw the wiring diagram using two

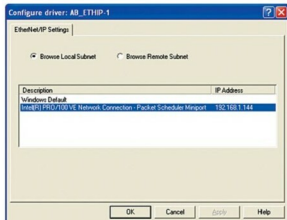


Fig. 6: Interfacing the PLC via the Ethernet to a PC to make it available to software for configuration and programming

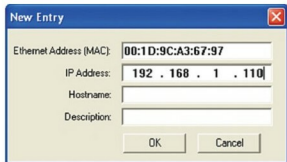


Fig. 7: Setting BOOTP server properties

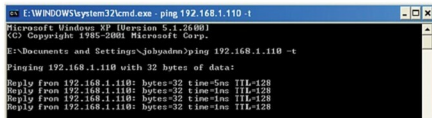


Fig. 8: Pinging to check if PLC connection is established

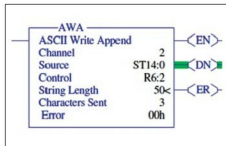


Fig. 9: AWA instruction set

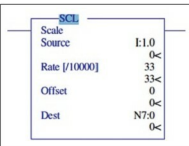


Fig. 10: SCL instruction set